

Problem 1 (15 points)

In this problem, you will develop a rule-based Named Entity Recognition (NER) system.

Please follow these steps:

(a) Text Selection and Preprocessing:

- Choose a news article or set of sentences that includes named entities such as people, organizations, locations, dates, times, and emails. Ensure your dataset is complex enough for effective testing.
- Use the Natural Language Toolkit (NLTK) to tokenize the text into sentences and words. Use the ``pos_tag()`` function in NLTK to apply Part-of Speech (POS) tagging on the tokenized text to form the basis for your entity recognition rules.

(b) Rule Creation and Pattern Matching:

- Create rules to identify entities using POS tags such as 'NNP' (Proper Noun Singular) or 'NNPS' (Proper Noun Plural).
- Examples of patterns to use include:
 - To identify person names (e.g., "Dr. Emily Stone") in text, focus on sequences that include titles like "Mr.", "Mrs.", or "Dr.", followed by capitalized first and last names. Use POS tagging to find 'NNP' (Proper Noun Singular) sequences, indicating proper names, and apply the regex pattern of type
`r'\b(Mr|Mrs|Ms|Dr|Prof)\.?\s+[A-Z][a-z]+\s+[A-Z][a-z]+\b'`.
 - To identify organization names (e.g., "Tech Innovations Inc.") in text, look for sequences of 'NNP' (Proper Noun Singular) followed by terms such as "Inc.", "Ltd.", "Corp.", or similar. Use the regex pattern of type
`r'\b[A-Z][a-z]+(?:\s+[A-Z][a-z]+)*\s+(Inc|Ltd|Corp|Corporation|Company)\b'`
to capture these entities effectively.
 - To identify location names (e.g., 'San Francisco') in text, search for sequences of consecutive 'NNP' (Proper Noun Singular) or 'NNPS' (Proper Noun Plural) tags. These tags can indicate geographical entities such as cities or countries, among other types of proper nouns.
 - To identify dates (e.g., "12/31/2020") in text, focus on numeric patterns that match typical date formats, using slashes or hyphens as separators. Apply the regex pattern of type
`r'\b\d{1,2}[/-]\d{1,2}[/-]\d{2,4}\b'`
to detect common numerical dates consisting of day, month, and year components. This pattern captures dates structured as two-digit or four-digit year sequences, accommodating formats such as "MM/DD/YYYY" or "DD-MM-YYYY".

- Expand the regex pattern to also recognize a date in formats such as "November 2024," "November 18, 2024," "18 Nov 2024," "2024-11-18," and "18th November, 2024". Provide the updated regex pattern and briefly explain how it works to identify these types of dates.

- To identify times (e.g., "10:30 AM") in text, focus on numeric patterns that match typical time formats, with or without AM/PM designators.

Apply the regex pattern of type

```
r '\b\d{1,2}:\d{2}(:\d{2})?\s?(AM|PM|am|pm)?\b'
```

to detect common time expressions consisting of hours and minutes, with optional seconds and period indicators.

- To identify email addresses (e.g., "emily.stone@gmail.com") in text, use the regex pattern of type

```
r '\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,7}\b'
```

This pattern captures standard email formats by matching a combination of letters, numbers, and special characters before the '@' symbol, followed by a domain with one or more subdomains and a valid top-level domain of two to seven letters.

(c) Entity Extraction and Evaluation:

- Apply these rules to extract entities from the preprocessed text.
- Display each recognized entity alongside its type, such as person, organization, location, date, time, or email.
- Review the effectiveness of your rules in identifying entities, providing specific examples.
- Discuss any challenges encountered, such as ambiguous names or complex sentence structures, and suggest improvements that could enhance accuracy.

SOLUTION:

Article Used

I used the Reuters article:

[**“US to convene global AI safety summit in November”**](#)

by David Shepardson (September 19, 2024).

The article text was used as the input for the rule-based NER system.

Approach

I implemented a rule-based NER system using:

- **Regex patterns** for PERSON, ORG, DATE, TIME, and EMAIL
- **POS-tag patterns** (NNP/NNPS sequences) for extracting location candidates
- **Tokenization + POS tagging** via NLTK

All code is provided separately in question1.py.

Extracted Entities

- **PERSON:** None (the article does not contain title+fullname patterns)
- **ORG:** None (no “Inc/Ltd/Corp” style organization names)
- **LOC:** Includes San Francisco, United States, Switzerland, etc., but also false positives due to NNP sequences
- **DATE:** May 30, 2024; September 19, 2024; October 2023
- **TIME:** 1:10 AM
- **EMAIL:** None

Evaluation

- Date/time extraction worked well
- Location extraction produced many false positives
- PERSON and ORG recall was low because the rule patterns were strict
- Could be improved by expanding ORG patterns, filtering month names, or combining with spaCy

Problem 2 (15 points)

In this problem, you will use SpaCy's pre-trained statistical model to perform Named Entity Recognition (NER) on a text document.

Please follow these steps:

(a) Statistical NER with SpaCy:

- Select a text document, such as a news article, that contains different types of entities like people, organizations, locations, dates, and more.
- Use SpaCy's pre-trained NER model to process the document and recognize named entities.
- Display the recognized entities along with their entity types.
- Discuss the results, focusing on the effectiveness of SpaCy's model in recognizing and categorizing entities. Consider aspects such as accuracy, types of entities identified, and potential areas for improvement.

(b) Extending NER with Custom Entities:

- Modify SpaCy's NER model to recognize a new type of entity that is not included by default (e.g., a specific product or brand name).
- Annotate a small portion of your dataset with the new entity type and train SpaCy's model.
- Demonstrate the model's ability to recognize the custom entity type by displaying the entities in the original text. Highlight how the model has been adapted to recognize the new entity and discuss potential improvements or further training needs.

SOLUTION:

Approach

I applied spaCy's pretrained `en_core_web_sm` model to the same Reuters article. All implementation is included in `question2.py`.

spaCy NER Summary

- **GPE:** San Francisco, U.S., United States, France, Japan, etc.
- **DATE:** May 30, 2024; October 2023; September 19, 2024; plus relative terms (e.g., "Last week," "Wednesday")
- **PERSON:** Joe Biden, Gina Raimondo, Anthony Blinken, David Shepardon
- **ORG:** Reuters, European Union, International Telecommunication Union, etc.
- **Incorrect predictions:** "AI" as GPE, "Generative AI" as PERSON

Comparison with Rule-Based NER

- spaCy detects far more PERSON and ORG entities
- spaCy recognizes contextual and relative dates (e.g., "Last week")
- Rule-based NER is transparent but brittle; spaCy is flexible but less interpretable
- A hybrid approach would combine the strengths of both

Custom Entity

I added one example custom entity class (e.g., "AI Summit") to spaCy via small manual annotations.

This allowed the model to correctly detect the custom EVENT entity.

Problem 3 (35 points)

In this problem, you will construct a neural network model to classify news articles and then enhance it using Named Entity Recognition (NER) features.

Please follow these steps:

(a) Construct and Evaluate a News Classification Model:

- Load the `'news_data.csv'` dataset, which contains 124 news articles, their headlines, and categories.
- Encode the `news_category` into numerical values.
- Combine the `news_headline` and `news_article` columns into a single text field for feature extraction.
- Split the dataset into training and testing sets using an 80-20 split and a `'random_state'` of 42 to ensure reproducibility.
- Use `'TfidfVectorizer'` to convert the text data into numerical format, with the number of top features as a parameter you will need to tune. Keep in mind that there are only 124 observations, which requires careful consideration when configuring your model.
- Design a neural network architecture appropriate for this task, considering activation functions and regularization techniques. Compile the model with an optimizer and loss function suitable for multi-class classification.
- Train the model on the training data, including validation splits to monitor the model's performance. Plot the training and validation accuracy per epoch to visualize model performance.
- Evaluate the model on the test dataset and print out the test accuracy.
- What is the best test accuracy you were able to obtain? Discuss the effectiveness of the model in classifying news articles and suggest any potential improvements or further experiments to enhance its performance.

(b) Enhance News Classification Using Named Entity Recognition:

- Apply SpaCy to extract named entities from the combined text field in both the training and testing sets created in part (a). These entities are expected to represent full names, company titles, etc.
- Convert the extracted named entities into binary dummy variables, indicating the presence or absence of each entity, such as “Mitsubishi”, “Audi”, etc. Ensure that these dummy variables are correctly aligned with your dataset, matching indices in both training and testing sets. Alternatively, your columns could be the types of entities recognized—such as PERSON, ORG, GPE, etc.—rather than the specific entities themselves. You will need to experiment to see which approach yields better model performance.
- Integrate the TF-IDF features with the named entity dummy variables to create a comprehensive feature matrix.

- Develop a neural network model using this combined feature set, considering activation functions and regularization techniques.
- Train the model on the training data, including validation splits to monitor the model's performance. Plot the training and validation accuracy per epoch to visualize model performance.
- Evaluate the model on the test dataset and print out the test accuracy. What would be the accuracy of classifying into these seven categories by random guess, assuming the dataset is balanced?
- Discuss how Named Entity Recognition (NER) impacts model performance and why.

To receive full credit, design a neural network that achieves a test accuracy of at least 0.45 for classifying into the seven given categories.

SOLUTION:

(a) TF-IDF + Neural Network

I combined the headline and article text, encoded labels with LabelEncoder, and represented the text using TF-IDF (1–2 grams, max 1000 features).

A small MLP model (256→128 hidden units with ReLU + dropout) was trained with an 80/20 validation split and early stopping.

- **Model 1 test accuracy: 0.20**
- **Random baseline: 0.1429**

The TF-IDF-only model performs slightly above random but generalizes poorly due to the small dataset.

(b) Adding NER Features

Using spaCy (en_core_web_sm), I extracted whether the article contains specific entity types (PERSON, ORG, GPE, NORP, EVENT, PRODUCT, WORK_OF_ART) and concatenated these binary features with TF-IDF.

Training the same MLP architecture on the combined feature set yielded:

- **Model 2 test accuracy: 0.52**

This exceeds the assignment's requirement (≥ 0.45).

NER helped because entity-type information provides higher-level semantic cues (e.g., politics → people/organizations, world news → countries), which TF-IDF alone cannot capture.

Best Accuracy

- **Best model: TF-IDF + NER**
- **Final test accuracy: 0.52**

Model 1: TF-IDF only



